

Capstone Project Integration or Webpage System: Node.js + MultiFile + MongoDB

Instructions:

Project Overview

This final project is a **group activity** that requires students to develop a web-based application integrating **Node.js**, **MultiFile project structure**, and **MongoDB**.

The project aims to apply the concepts and skills learned in web development by creating a functional system that demonstrates backend programming, file organization, and database connectivity using MongoDB as a NoSQL database.

Students are expected to collaborate as a team, design the system architecture, implement the application, and present their completed project.

Project Objectives

- At the end of the project, students should be able to:
1. Develop a functional webpage application using **Node.js**.
 2. Apply **MultiFile project organization** for better code management and maintainability.
 3. Integrate **MongoDB** as a NoSQL database for storing and retrieving information.
 4. Demonstrate teamwork, system development, and technical presentation skills.
 5. Apply web development concepts in creating a complete capstone project.

Project Requirements

1. Capstone Project Integration or Webpage System

The project must integrate the following technologies:

Frontend: • HTML • CSS • JavaScript	Backend: • Node.js • Express.js	Database: • MongoDB (NoSQL Database)
--	--	---

2. MultiFile Project Structure

The project must follow a MultiFile structure to organize system components properly.

Suggested structure:

```
ProjectName/  
  
├── server.js  
├── package.json  
  
├── routes/  
|   └── routes.js  
  
├── models/  
|   └── databaseModel.js  
  
└── views/
```

| └─ webpage files

| └─ public/

| └─ css/

| └─ javascript/

└─ database/

Students may modify the structure depending on their project requirements.



Group Activity Guidelines

- The project must be completed by a group.
- Each member should have an assigned role and contribution.
- Groups are responsible for:
 - Planning the project
 - Developing the system
 - Testing the application
 - Preparing documentation and presentation
- Suggested group roles:
 - Project Leader
 - Backend Developer
 - Frontend Developer
 - Database Administrator
 - Documentation and Presentation Coordinator



Project Timeline

Date	Activities	Expected Output	Deliverable
Task 1 July 9, 2026	Project Planning and System Design	Project Proposal Documentation UI Design Screenshots (prototype)	
Task 2 July 15 and 17 2026	Development and Integration	Node.js setup, MultiFile implementation, MongoDB connection, and development of system features	
Task 3 July 20, 2026	Finalization and Presentation Preparation	System testing, completion of project files, preparation of presentation slides, and final submission	
Task 4 July 24, 2026	Final Submission	Complete Project Requirements	Laboratory: Submit the complete video presentation of the project , demonstrating the project's objectives, design, implementation, key

Date	Activities	Expected Output	Deliverable
			features, system functionality, and results. Lecture: Submit a printed copy of the documentation and upload the PDF version to the Schoolbook platform on or before the specified deadline.

Lecture Deliverable

Project Presentation

Each group must submit a recorded presentation and a presentation document.

Presentation Requirements:

- 3 minutes minimum and 5 minutes maximum recording duration
- PowerPoint presentation converted into **PDF format**
- Total number of slides: **7 to 10 slides**

Required Presentation Structure (7 Slides)

Slide 1: Title Page

- Project Title
- Group Members with photo
- Course and Section

Slide 2: Project Introduction

- Background of the project
- Purpose of the system

Slide 3: Objectives and System Overview

- Project objectives
- Overview of the developed application

Slide 4: System Architecture

- Frontend, Backend, and Database integration

- Node.js + MultiFile + MongoDB workflow

Slide 5: Database Design

- MongoDB database structure
- Sample database records

Slide 6: System Features and Demonstration

- Main functionalities
- Screenshots of the developed system

Slide 7: Conclusion and Future Improvements

- Summary of the project
- Possible enhancements

RUBRICS

Criteria	Percentage
System Functionality and Implementation	40%
Node.js and MultiFile Integration	25%
MongoDB Database Integration	20%
Presentation Quality and Documentation	10%
Group Collaboration	5%

Criteria	Percentage
Total	100%

Note: The final project should demonstrate the successful integration of **Node.js, multi-file application development, and MongoDB database management** through a functional web-based application.